

Jay Bombatkar

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Objective

Electronics and Telecommunication Engineering final-year student with hands-on experience in embedded systems, PCB design, and microcontroller-based hardware development. Seeking an entry-level role in Embedded Systems, Hardware Design, or Electronics R&D to apply technical skills in real-world product development.

Education

Bachelor of Engineering in Electronics and Telecommunication 2022 – 2026
Savitribai Phule Pune University

- SGPA : 8.85

Higher Secondary School, 2020 – 2022

- Percentage : 70.83

Experience

Intern – SuryaLogix Pvt. Ltd. Feb 2026 – April 2026

- Performed testing of environmental and solar sensors (temperature, humidity, irradiation) to validate accuracy and performance
- Conducted PCB-level testing and debugging to identify faults in sensor circuits and ensure reliable operation
- Verified sensor outputs (analog/digital, RS-485) under different operating conditions
- Assisted in production testing and quality checks to ensure compliance with industry standards
- Collaborated with team for issue analysis, troubleshooting, and process improvement

Skills

Programming	: Python, Embedded C, SQL
Embedded Systems	: Microcontrollers (PIC, STM32), Interrupts, Timer Programming, Memory Interfacing
Communication Protocols	: UART, SPI, I2C, USB, CAN Bus
Debugging tools	: Oscilloscope, Multimeter
Tools & Concepts	: Git, GitHub, SDLC, V-Model, FreeRTOS, Linux
PCB Design	: KiCad, Altium (schematic design, footprint assignment, PCB layout, routing, multi-layer)
Simulation	: LTspice (circuit simulation, waveform analysis)
Development Tools	: MPLAB X IDE, MPLAB IPE, CCS Programmer
Core Coursework	: Analog Electronics, Digital Electronics, Microcontrollers, Communication Systems
Additional Skills	: Datasheet Analysis, Component Selection, Hardware Validation, Circuit Troubleshooting
Certifications	: ◦ PCB Design with KiCad (Udemy) ◦ Hardware Design using PIC Microcontroller (Udemy)

Projects

- **CNC Controller Using MPG (Manual Pulse Generator)** *Ongoing*
Developed a microcontroller-based CNC jogging controller for precise manual control of X, Y, and Z axes using an MPG encoder.
Designed and validated 2-layer PCB including schematic capture, routing, grounding, and EMI considerations.
Implemented embedded C firmware for encoder interfacing, pulse generation, debouncing, and real-time control.
Performed hardware testing and debugging with oscilloscope and MPLAB to ensure accurate and reliable system performance.
- **STM32-Based Embedded System PCB Design (Altium)**
Designed a complete STM32-based embedded system PCB using Altium Designer, including schematic capture and PCB layout.
Implemented power supply circuitry, microcontroller interfacing, and communication interfaces (USB/UART).
Performed component placement, routing, and design rule checks (DRC) to ensure signal integrity and manufacturability.
Generated fabrication files (Gerber, BOM).

Achievements & Leadership

- Lead in Project Expo 2026 event at PVPIT, Bavdhan
- Lead in Quiz Competition 2024 event at PVPIT, Bavdhan